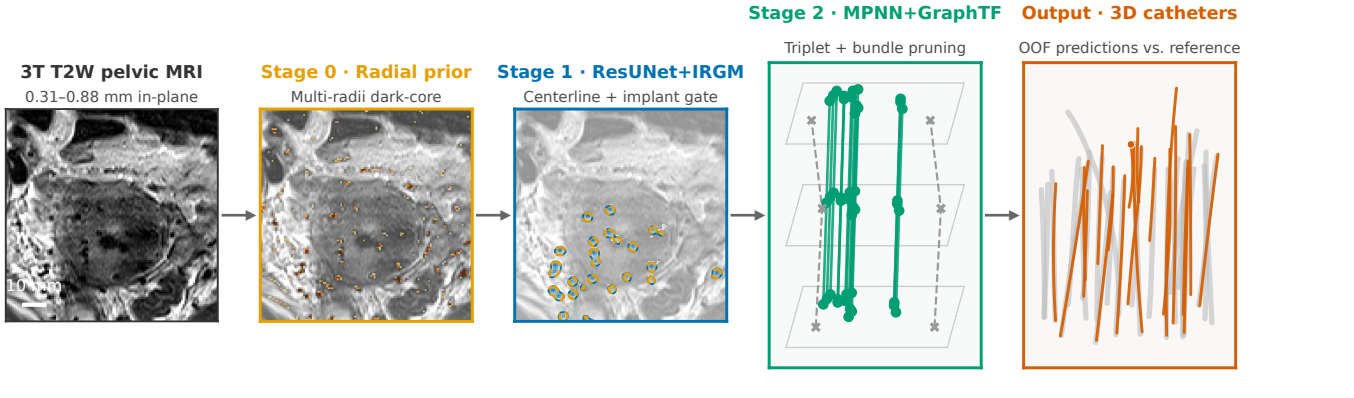
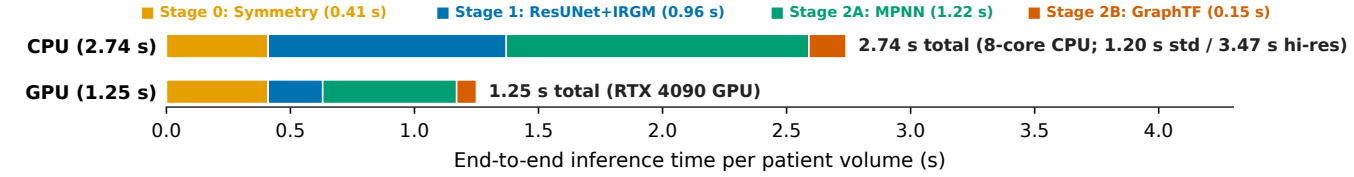

3T T2W pelvic MRI
0.31–0.88 mm in-plane
10 mm
Stage 0 · Radial prior
Multi-radii dark-core
Stage 1 · ResUNet+IRGM
Centerline + implant gate
Stage 2 · MPNN+GraphTF
Triplet + bundle pruning
Output · 3D catheters
OOF predictions vs. reference


Stage 0: Symmetry (0.41 s)
Stage 1: ResUNet+IRGM (0.96 s)
Stage 2A: MPNN (1.22 s)
Stage 2B: GraphTF (0.15 s)
CPU (2.74 s)
2.74 s total (8-core CPU; 1.20 s std / 3.47 s hi-res)
GPU (1.25 s)
1.25 s total (RTX 4090 GPU)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
End-to-end inference time per patient volume (s)